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**UNIVERSITY OF LIMPOPO
FACULTY OF HUMANITIES
DEPARTMENT OF MATHEMATICS, SCIENCES AND TECHNOLOGY
EDUCATION**

**LEARNERS' COMPUTATION OF TRIGONOMETRY 2D PROBLEMS
USING PROBLEM-BASED LEARNING AT A SCHOOL IN LEPELLE 04
CIRCUIT, MOPANI WEST DISTRICT, LIMPOPO PROVINCE, SOUTH
AFRICA**

A Research Proposal submitted in partial fulfilment of the requirements
for the degree of

HONOURS IN MATHEMATICS EDUCATION

by

MR MD [REDACTED]

Student Number: [REDACTED]

Supervisor: Prof ZB Dhlamini

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CHAPTER 1: BACKGROUND AND ORIENTATION OF THE STUDY

1.1 Introduction

Mathematics occupies a central place in daily life and in formal education because it equips individuals with the capacity to reason analytically, interpret quantitative information, and arrive at sound decisions across a wide range of contexts (Ngcobo et al., 2019). Within the school mathematics curriculum, trigonometry stands out as a strand that bridges numerical computation and geometric reasoning. The subject matter centres on the systematic relationships between the angles and sides of triangles, with the sine, cosine, and tangent ratios serving as the primary tools for quantifying those relationships (Department of Basic Education [DBE], 2011). At Grade 10 level, learners encounter formal trigonometry for the first time through the lens of two-dimensional (2D) right-angled triangle problems, and the Curriculum and Assessment Policy Statement (CAPS) requires them to calculate unknown sides and angles while connecting their work to authentic, real-life situations (DBE, 2011).

The importance of trigonometry extends well beyond the mathematics classroom. Mastery of trigonometric principles underpins a range of professions and fields, including civil and electrical engineering, architectural design, land surveying, navigation systems, and observational astronomy. At a foundational level, the topic cultivates problem-solving fluency and develops learners' capacity to translate abstract mathematical notation into workable procedures — competencies that are valued across the curriculum. Despite these acknowledged benefits, substantial evidence indicates that many Grade 10 learners in South African schools struggle to perform competently in trigonometry tasks. The challenges are multifaceted: learners frequently have difficulty distinguishing between trigonometric ratios, translating verbal or diagrammatic information into algebraic expressions, executing multi-step calculations accurately, and encoding their solutions in forms that convey mathematical meaning (Dhungana et al., 2023). Because trigonometry is introduced in Grade 10 without

significant prior scaffolding, learners who do not consolidate the conceptual foundations at this stage tend to carry their misconceptions forward into Grades 11 and 12, where the demands of the curriculum escalate considerably.

1.2 Background to the Study

Empirical evidence from South Africa and from comparable schooling systems internationally confirms that trigonometry constitutes one of the most persistently challenging domains within secondary school mathematics. Msi et al. (2026) conducted a study with Grade 10 learners in a rural Eastern Cape school and identified a pattern of errors spanning the comprehension, transformation, processing, and encoding stages of problem solving. Their findings revealed that learners routinely misread the structural relationships encoded in trigonometric diagrams, applied incorrect identities during transformation, and produced solutions that did not accurately reflect the reasoning steps undertaken. Adebayo (2023) similarly found that Grade 11 learners working with trigonometric function graphs manifested deep-seated misconceptions about the behaviour of sine, cosine, and tangent across different quadrants, and that these misconceptions were compounded by inadequate algebraic manipulation skills. At a macro level, Ndebele (2024) analysed National Senior Certificate (NSC) performance data and attributed persistently low scores in the trigonometry section of Mathematics Paper 2 to entrenched learner errors and to teachers' limited capacity to diagnose and remediate those errors systematically.

The specific types of errors that learners make when working through trigonometric problems have been documented across multiple studies. Sahlu et al. (2025) found that secondary school learners in an Ethiopian context experienced greatest difficulty at the comprehension and numerical calculation stages, reflecting an inability to extract the relevant mathematical information from problem prompts and to carry out arithmetic procedures without error. Arhin and Hokor (2021) reported that many secondary school learners lack

robust conceptual understanding of what trigonometric ratios actually represent, relying instead on surface-level pattern matching that breaks down as soon as problems deviate from familiar formats. Ngu and Phan (2023) demonstrated that the challenge is intensified when problems require learners to perform algebraic transformations — for instance, rearranging a trigonometric equation to isolate an unknown angle or side — because such tasks demand the simultaneous coordination of trigonometric reasoning and algebraic manipulation. Taken together, these findings indicate that the difficulties learners encounter in trigonometry are not incidental or idiosyncratic; they reflect systematic gaps in both conceptual grounding and procedural fluency.

62 In the South African context, research has further highlighted the role of pedagogical factors in shaping learner performance. Luneta (2015) argued that many South African learners underperform in mathematical problem solving partly because classroom instruction emphasises memorisation and routine drill at the expense of conceptual exploration. When learners are expected to reproduce procedures without understanding the principles that justify them, they become unable to adapt their knowledge to novel problem configurations — a limitation that is especially consequential in trigonometry, where problems are rarely presented in a single standard format. Human (2009) made a related observation, noting that teacher-centred approaches position learners as passive recipients of declarative knowledge rather than as active constructors of understanding, thereby limiting opportunities for the kind of exploratory engagement that promotes durable learning.

56 The foregoing evidence suggests a clear need for pedagogical intervention at the point where trigonometry is formally introduced. A teaching approach that creates structured opportunities for learners to engage collaboratively with authentic, contextualised problems has the potential to address both the conceptual and procedural dimensions of learner difficulty. Problem-Based Learning (PBL) is one such approach. Defined as a learner-centred

instructional strategy in which groups of learners work collaboratively to resolve meaningful, ill-structured problems while the teacher assumes a facilitative rather than a directive role, PBL has been associated with improvements in critical thinking, conceptual understanding, collaborative skill development, and intrinsic motivation across a range of subject areas (Yew & Goh, 2016). Amedume et al. (2022) found that the introduction of PBL into mathematics classrooms resulted in heightened learner participation and marked gains in both content understanding and independent problem-solving ability. Barco Rojas et al. (2022) reached similar conclusions in a systematic review, concluding that PBL produces measurable improvements in mathematical achievement when implemented with fidelity to its core design principles.

1.3 Statement of the Research Problem

In spite of the centrality of trigonometry to the Grade 10 mathematics curriculum, many learners enrolled in schools within the Lepelle 04 Circuit of Mopani West District continue to perform poorly on trigonometry 2D tasks. Classroom observation and assessment data indicate that learners regularly commit errors at each of the four stages of mathematical problem solving — comprehension, transformation, processing, and encoding — when working on tasks that require the application of the sine, cosine, and tangent ratios to right-angled triangle problems.

These difficulties are not attributable to a single cause. Rather, they reflect the confluence of underdeveloped algebraic foundations, limited diagram literacy, a tendency to rely on rote memorisation rather than conceptual understanding, and exposure to predominantly teacher-centred instructional approaches that afford learners few opportunities to negotiate meaning collaboratively.

Although a body of research has examined learner difficulties in trigonometry and has explored various learner-centred pedagogies in mathematics education more broadly, comparatively little attention has been directed specifically toward the potential of Problem-

Based Learning to improve Grade 10 learners' computation of 2D trigonometry problems in rural school settings within Limpopo Province. Most available studies address Grade 11 or Grade 12 learners, or they investigate PBL in mathematical domains other than trigonometry.

The present study therefore seeks to address this gap by investigating how the systematic integration of PBL into the teaching of Grade 10 trigonometry can support learners in developing more accurate and more flexible computational skills in the 2D domain.

1.4 Purpose and Research Questions

The purpose of this study is to explore how Problem-Based Learning can support Grade 10 mathematics learners in computing two-dimensional trigonometry problems at a school in Lepelle 04 Circuit, Mopani West District, Limpopo Province, South Africa.

The study is guided by the following research questions:

RQ1. What challenges do Grade 10 learners experience when computing two-dimensional trigonometry problems?

RQ2. How does the integration of Problem-Based Learning influence Grade 10 learners' computation of two-dimensional trigonometry problems?

1.5 Significance of the Study

This study holds significance at three levels. Practically, the findings may assist mathematics teachers in the Mopani West District and in analogous rural districts to design and implement PBL activities that are responsive to the specific computational challenges learners encounter in trigonometry. Curriculum advisors and school management teams may draw on the study's findings to inform professional development programmes and to advocate for the adoption of learner-centred pedagogies that are aligned with the vision of CAPS. For learners in rural and previously disadvantaged schools, the study offers the prospect of a more engaging

and more effective instructional experience in a topic that has historically been a significant barrier to mathematical progression (Stott & Duvenhage, 2024).

Theoretically, the study extends the existing knowledge base on the application of Constructivist Learning Theory and Problem-Based Learning within the South African mathematics education context, with a particular focus on a grade level and a circuit that have received limited scholarly attention. By generating detailed, contextualised evidence on the relationship between PBL and trigonometric computation, the study contributes to a more nuanced understanding of how learner-centred approaches can be tailored to the demands of specific content areas and specific school environments (Maphutha et al., 2023).

1.6 Delimitations and Scope

This study is confined to one secondary school in the Lepelle 04 Circuit of Mopani West District, Limpopo Province. The focus is on Grade 10 mathematics learners and their engagement with two-dimensional trigonometry problems during a defined period of instruction conducted through PBL. The study does not extend to three-dimensional trigonometry, to other topics within the Grade 10 mathematics curriculum, or to other grade levels. The qualitative nature of the inquiry means that statistical generalisation to the broader population of Grade 10 learners is not sought; rather, the study aims to generate rich, contextualised insights that may be transferable to comparable settings.

1.7 Overview of the Proposal

Chapter 1 has provided the background, motivation, research problem, and research questions that frame the study. Chapter 2 presents a critical review of the empirical literature pertaining to Grade 10 learners' challenges in trigonometry and to the use of Problem-Based Learning in mathematics education, culminating in an articulation of the theoretical framework that underpins the research design. Chapter 3 details the research methodology, including the

research paradigm, design, sampling strategy, data collection instruments, data analysis procedures, and quality criteria. The proposal concludes with a discussion of ethical considerations, a reference list, and supporting appendices.

CHAPTER 2: LITERATURE REVIEW AND THEORETICAL FRAMEWORK

2.1 Introduction

This chapter critically reviews the empirical literature relevant to two interconnected areas of enquiry: (a) the challenges that Grade 10 learners encounter when learning and computing trigonometry 2D problems, and (b) the documented effects of Problem-Based Learning on mathematical understanding and performance. The chapter also presents the theoretical framework — grounded in Constructivist Learning Theory and its Social Constructivist extension — that guides the epistemological and pedagogical assumptions of the study. The review is organised thematically rather than chronologically, with the aim of identifying convergences and divergences across studies and of foregrounding the specific empirical gap that this investigation seeks to address.

2.2 Grade 10 Learners' Challenges in Trigonometry

2.2.1 Error Patterns and Misconceptions

A substantial body of empirical evidence documents the types of errors and misconceptions that secondary school learners manifest when engaging with trigonometric problems. Newman's Error Analysis framework, which categorises errors as comprehension, transformation, processing, encoding, or careless, has been widely applied in South African and international studies and consistently yields evidence of difficulties distributed across all categories. Msi et al. (2026) applied this framework to the written work of Grade 10 learners in a rural Eastern Cape school and found that comprehension errors — specifically, learners' inability to identify the correct trigonometric ratio from a diagrammatic prompt — were the most prevalent category. Transformation errors, arising when learners correctly comprehend the problem but select an inappropriate mathematical procedure, were the second most common type. Encoding errors, in which learners execute correct procedures but fail to express

their answers in the required form, were also recorded, suggesting that difficulties are not confined to the conceptual domain but extend to communicative competence.

Sahlu et al. (2025) replicated and extended these findings in their analysis of Grade 10 learners' trigonometry scripts from an Ethiopian secondary school context, noting that numerical calculation errors — particularly errors arising from incorrect calculator use and from confusion between degree and radian modes — constituted a significant proportion of the total error count. This finding resonates with the observations of Dhungana et al. (2023), who found that learners in Nepal's secondary schools were frequently unable to distinguish between the conceptual meaning of a trigonometric ratio and the procedural steps required to evaluate it. Although these studies were conducted in different national contexts, the convergence of their findings points to a set of challenges that are not culturally specific but that reflect the inherent cognitive demands of trigonometric problem solving.

South African studies add an important contextual dimension to this picture. Adebayo (2023) found that Grade 11 learners' difficulties with trigonometric function graphs were exacerbated by inadequate conceptual foundations established during Grade 10 instruction, and argued that errors in the function graph context were often traceable to persistent misconceptions about the unit circle that were never adequately addressed at the introductory level. Ndebele (2024) reached a similar conclusion through an analysis of NSC examination performance, contending that the sustained underperformance of South African learners in the trigonometry component of Mathematics Paper 2 reflects a systemic failure to develop robust conceptual understanding rather than merely a deficit in procedural skill. Mutodi and Maphutha (2023) added nuance to this argument by demonstrating that learners' graphical interpretation difficulties in trigonometry are compounded by limited spatial reasoning ability, which makes it difficult for them to visualise the geometric relationships that the trigonometric ratios express.

2.2.2 Algebraic Foundations and Procedural Fluency

Multiple studies have identified weak algebraic foundations as a proximate cause of poor trigonometric performance. Ngu and Phan (2023) conducted an experimental study comparing different instructional formats and found that learners' ability to solve trigonometry problems involving algebraic transformation — for instance, rearranging a ratio equation to express an unknown angle — was strongly predicted by their existing algebraic proficiency. Learners who lacked facility with inverse operations, equation balancing, and symbolic manipulation were significantly more likely to commit transformation and processing errors on trigonometric tasks. Maphutha et al. (2023) reached a related conclusion in their investigation of Grade 11 learners' mathematical connections in activity-based learning environments, observing that learners who could not fluidly apply algebraic reasoning were unable to exploit the interconnected structure of two-dimensional trigonometric problems and therefore relied on fragmented, step-by-step recall rather than coherent problem-solving strategies.

Luneta (2015) argued that the prevalence of algebraic deficits among South African secondary school learners is partly attributable to the memorisation-centred approaches that dominate classroom practice, which result in learners acquiring surface-level familiarity with procedures without understanding the underlying mathematical structures. This line of argument is consistent with the broader critique of teacher-centred pedagogy advanced by Human (2009), who contended that passive reception of mathematical knowledge is fundamentally incompatible with the development of the flexible, transferable understanding that rigorous problem solving demands. Arhin and Hokor (2021) provided supporting evidence from the Ghanaian context, finding that learners who had been taught trigonometry through largely expository methods demonstrated significantly weaker conceptual understanding of trigonometric ratios than did learners who had participated in structured inquiry activities.

2.2.3 The South African Classroom Context

In many South African schools, particularly those serving rural communities, mathematics instruction continues to be characterised by large class sizes, limited access to technological resources, and a reliance on teacher-directed transmission of content (Stott & Duvenhage, 2024). These structural factors constrain the range of pedagogical approaches that teachers can realistically implement and create conditions in which learners receive limited opportunity for collaborative exploration or self-directed problem solving. The consequence, as Chigonga and Mutodi (2023) noted in a review of the South African mathematics education literature, is that learners' development of mathematical reasoning, analytical thinking, and communication skills is systematically impeded. This observation is particularly pertinent to the learning of trigonometry, which requires learners not only to execute procedures accurately but also to reason about the geometric and algebraic relationships that those procedures express.

2.3 Problem-Based Learning in Mathematics Education

2.3.1 Conceptual Foundations and Design Principles

Problem-Based Learning is a pedagogical approach that was originally developed in medical education and subsequently adapted for use across a wide range of disciplines, including mathematics. In its canonical form, PBL presents learners with a complex, authentic, and typically ill-structured problem as the starting point for learning, and requires them to work in small collaborative groups to identify what they know, determine what they need to know, and construct solutions through a cycle of investigation, discussion, and reflection (Savery, 2006). The teacher's role in a PBL environment is fundamentally different from that in a conventional lesson: rather than serving as the primary source of knowledge, the teacher acts as a facilitator who poses guiding questions, monitors group dynamics, and scaffolds learners' progress without pre-empting their reasoning. Yew and Goh (2016) identified four core

outcomes of well-implemented PBL: the development of critical thinking, the deepening of conceptual understanding, the cultivation of collaborative competency, and the promotion of self-directed learning. These outcomes are precisely those that the empirical literature identifies as being deficient in learners who struggle with trigonometric computation.

2.3.2 Empirical Evidence from Mathematics Education

Empirical studies investigating the effects of PBL on mathematical achievement have produced largely positive findings, although the magnitude of the effects varies across contexts and grade levels. Istian (2024) conducted a systematic literature review of PBL in mathematics courses and found consistent evidence that learners taught through PBL outperformed their counterparts in conventional instruction on measures of problem-solving skill, content achievement, and motivation. The review noted that the benefits of PBL were most pronounced for tasks that required learners to apply mathematical knowledge to novel situations, which aligns with the demands of trigonometry 2D problem solving. Amedume et al. (2022) investigated the impact of PBL on mathematics achievement in a Ghanaian secondary school context and found that learners in the PBL condition demonstrated significantly greater gains in content understanding, creative problem solving, and independent mathematical reasoning than did learners in the comparison group.

Sallah et al. (2023) conducted a study focused specifically on the relationship between PBL and error minimisation in trigonometry, and found that learners who participated in PBL-based trigonometry lessons committed fewer comprehension and transformation errors on post-instructional assessments than did learners taught through conventional methods. Critically, the PBL learners also demonstrated stronger procedural understanding, suggesting that the approach was effective not only in promoting conceptual insight but also in building the accuracy and fluency required for computation. Barco Rojas et al. (2022) synthesised the evidence from multiple studies and concluded that PBL in mathematics education improves

active participation, communication skills, and long-term retention of mathematical concepts, particularly when the problems used are genuinely contextualised and when group processes are adequately facilitated.

The use of manipulatives and activity-based learning resources has been shown to amplify the benefits of PBL in mathematics, particularly for topics that have a strong geometric or spatial dimension. Brijlall (2015) found that the use of physical manipulatives in a South African trigonometry lesson supported learners in constructing mental representations of angular relationships that they had previously been unable to visualise, and attributed this effect to the concrete, embodied nature of the learning experience. Niranjana and Brijlall (2025) extended this finding by demonstrating that a constructivist approach to trigonometry problem solving, incorporating hands-on tools and guided inquiry, produced measurable improvements in learners' spatial reasoning and computation skills in South African secondary schools. Maphutha et al. (2023) likewise found that collaborative, activity-based learning environments in which Grade 11 learners worked in groups to investigate trigonometric relationships produced stronger mathematical connections than did individual, textbook-based instruction.

2.3.3 Challenges in the Implementation of PBL

Despite the broadly positive evidence base, several studies have identified obstacles to the effective implementation of PBL in secondary school mathematics contexts. Vivic (2023) noted that teachers who have been trained primarily in teacher-centred approaches may find the facilitative role required by PBL unfamiliar and uncomfortable, and that without targeted professional development, PBL lessons can devolve into unstructured group activities that fail to produce the intended cognitive engagement. Time pressure is a related concern: PBL lessons are inherently more time-consuming than conventional lessons because learners require space to explore, discuss, and revise their thinking, and the demands of curriculum coverage may lead teachers to curtail the problem-solving process before learners have reached adequate

understanding. In resource-constrained schools, the design of authentic, well-structured problems that are accessible to learners with diverse levels of mathematical readiness presents an additional challenge. Chigonga and Mutodi (2023) emphasised that effective PBL in mathematics requires careful problem design that aligns the level of challenge with learners' zone of proximal development, ensuring that problems are neither so simple as to be trivial nor so complex as to produce cognitive overload.

Research further indicates that some learners, particularly those who have been socialised into passive learning roles through years of teacher-centred instruction, may initially resist the demands of PBL and may require sustained scaffolding before they are able to engage productively with collaborative problem-solving activities. Stott and Duvenhage (2024) found that higher-achieving South African learners from socioeconomically disadvantaged backgrounds responded positively to mathematics-guided PBL programmes once they had been oriented to the expectations of the approach, suggesting that the initial adjustment period can be navigated successfully when teachers provide adequate support. These findings collectively indicate that the implementation of PBL is not a panacea but requires deliberate planning, contextual sensitivity, and ongoing formative assessment if it is to realise its potential as a lever for improving learner performance in trigonometry.

2.4 Research Gap

The literature reviewed above establishes a well-documented case for the persistence of trigonometric computation difficulties among Grade 10 learners and for the potential of Problem-Based Learning to address those difficulties. However, a significant gap remains in the research base. The majority of studies examining learner difficulties in trigonometry focus on Grade 11 or Grade 12, where learners have already had at least one year of exposure to the topic; comparatively few studies have examined the formative period of Grade 10 instruction, when the conceptual foundations for all subsequent trigonometric learning are established.

Moreover, while the effectiveness of PBL in mathematics has been investigated in various international and South African contexts, there is a notable absence of empirical evidence focused specifically on the use of PBL to improve Grade 10 learners' computation of 2D trigonometry problems in rural schools within Limpopo Province. This study is designed to address that gap.

2.5 Theoretical Framework

2.5.1 Constructivist Learning Theory

The study is grounded in Constructivist Learning Theory, which holds that knowledge is not passively received from an external source but is actively constructed by the learner through processes of interaction, exploration, and reflection (Piaget, 1970). From a constructivist perspective, the role of instruction is not to transmit pre-formed knowledge to a receptive learner but to create conditions in which learners can engage with challenging material, encounter cognitive conflict, and progressively reorganise their conceptual structures in the direction of more sophisticated and more coherent understanding. Piaget's (1970) concepts of assimilation and accommodation are directly relevant to the learning of trigonometry: when learners encounter a trigonometric problem that does not fit their existing cognitive schema, they must either assimilate the new information into a modified version of that schema or accommodate their schema to incorporate the new knowledge. PBL, with its emphasis on authentic, challenging problems, creates repeated opportunities for both processes.

Constructivist theory has been extensively applied in contemporary South African mathematics education research. Niranjana and Brijlall (2025) drew on constructivist principles to design a trigonometry intervention in which learners constructed understanding of angular relationships through hands-on exploration, and found that the intervention produced significant gains in both procedural accuracy and conceptual depth. Maphutha et al. (2023)

similarly applied constructivist ideas to analyse Grade 11 learners' mathematical connections in an activity-based learning environment, finding that learners who were given opportunities to construct knowledge collaboratively demonstrated more robust and more transferable understanding than did their counterparts in conventional instructional settings.

2.5.2 Social Constructivism and the Zone of Proximal Development

The study additionally draws on Vygotsky's (1978) Social Constructivism, which extends Piagetian constructivism by foregrounding the role of social interaction in the construction of knowledge. Vygotsky argued that higher-order cognitive functions — including mathematical reasoning, analytical thinking, and problem-solving — develop first in the social plane, through collaborative activity with more knowledgeable others, and are subsequently internalised by the individual learner. Central to this theoretical position is the concept of the Zone of Proximal Development (ZPD), defined as the distance between what a learner can accomplish independently and what the learner can accomplish with appropriate guidance and collaboration. Within the ZPD, learning is optimised when tasks are calibrated at a level of challenge that exceeds the learner's independent capability but remains achievable with support.

In the context of this study, the ZPD concept has direct implications for the design and facilitation of PBL activities. When Grade 10 learners work in small groups to solve 2D trigonometry problems, the group context provides a micro-social space in which more mathematically confident learners can scaffold the reasoning of their peers, and in which the teacher can intervene strategically to extend learners' thinking beyond what they would achieve individually. Vygotsky's (1978) theory therefore provides a principled justification for the collaborative structure of PBL and for the facilitative role that the teacher assumes in a PBL lesson. Recent applications of Social Constructivism in South African mathematics education include the work of Chigonga and Mutodi (2023), who demonstrated that problem-solving

approaches grounded in social constructivist principles fostered the development of mathematical communication skills among secondary school learners, and the work of Stott and Duvenhage (2024), who found that structured collaborative learning enhanced both mathematical reasoning and academic self-efficacy in learners from disadvantaged school contexts.

2.5.3 Alignment Between Theoretical Framework and PBL

The alignment between Constructivist Learning Theory, Social Constructivism, and Problem-Based Learning is both conceptual and practical. Conceptually, all three frameworks share the core commitment to positioning the learner as an active agent in the construction of knowledge and to viewing the social environment as a constitutive rather than a merely supportive dimension of the learning process. Practically, the design features of PBL — collaborative group work, authentic problem contexts, facilitative teaching, and iterative cycles of inquiry and reflection — operationalise the key principles of Constructivist and Social Constructivist theory in the classroom. In the specific context of Grade 10 trigonometry, this alignment means that PBL lessons can be designed to create the conditions in which learners encounter cognitively stimulating problems, negotiate meaning with their peers within the ZPD, and progressively construct the conceptual and procedural knowledge needed to compute 2D trigonometry problems accurately and flexibly. The theoretical framework thus provides both a rationale for the pedagogical approach under investigation and a lens through which the data collected in the study can be interpreted.

2.6 Conclusion

This chapter has reviewed the empirical evidence on Grade 10 learners' challenges in trigonometry, established the potential of Problem-Based Learning as an instructional response to those challenges, and articulated the Constructivist and Social Constructivist theoretical framework that guides the study. The review has identified a clear empirical gap — the absence

of studies examining PBL in the context of Grade 10 trigonometry 2D computation in rural Limpopo Province — which the present research is designed to fill. Chapter 3 will outline the methodological approach through which the study will generate evidence relevant to this gap.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Introduction

This chapter presents and justifies the methodological choices that will guide the collection, analysis, and interpretation of data in this study. The chapter begins with a discussion of the research paradigm and design, proceeds to describe the sampling strategy, data collection instruments, and data analysis procedures, and concludes with a detailed account of the quality criteria and ethical considerations that will govern the conduct of the research.

3.2 Research Paradigm

The study is located within the interpretivist paradigm, which holds that social realities are multiple, constructed, and context-dependent, and that the goal of research is to understand the meanings that participants attach to their experiences rather than to discover universal laws that hold across contexts (Creswell & Creswell, 2018). The interpretivist paradigm is appropriate for this study because the research aims to understand how Grade 10 mathematics learners in a specific school context experience the learning of 2D trigonometry through PBL — a phenomenon that is inherently subjective, socially situated, and context-sensitive. An interpretivist approach permits the researcher to attend to the richness and complexity of learners' experiences and to generate nuanced insights that would be flattened by a purely quantitative methodology.

3.3 Research Design

The study will employ a qualitative case study research design. Merriam (2009) described the case study as a form of qualitative inquiry in which the researcher conducts an in-depth investigation of a bounded system — a person, a group, an event, or a process — in its natural context, using multiple sources of evidence to develop a rich, holistic understanding of the case. This design is well suited to the present study because it allows the researcher to

examine Grade 10 learners' computational challenges and problem-solving experiences in the particular context of one secondary school in Lepelle 04 Circuit, without imposing artificial conditions that would distort the naturalistic dynamics of the classroom.

The case study design is also consonant with the interpretivist paradigm adopted for this study, because it foregrounds contextual particularity and prioritises depth of understanding over breadth of coverage. By focusing on a single school as the bounded case, the researcher can draw on multiple data sources — classroom observations, semi-structured interviews, and document analysis — to construct a layered account of the phenomenon under investigation. The use of multiple sources of evidence is a defining strength of the case study design, as it enables triangulation and thereby enhances the credibility of the findings (Merriam, 2009).

3.4 Research Site and Sampling

The study will be conducted at a secondary school in the Lepelle 04 Circuit, Mopani West District, Limpopo Province, South Africa. The school serves a predominantly rural community, and the majority of its learners come from socioeconomically disadvantaged households. The school has been selected because it is representative of the type of context in which trigonometry underperformance has been documented, and because the researcher has established access to the school through formal channels.

Participants will be selected through purposive sampling, a non-probability technique in which the researcher deliberately selects individuals who possess the knowledge, experience, and characteristics required to address the research questions (Cohen et al., 2018). The primary participants will be Grade 10 mathematics learners who are enrolled in the school and who are studying 2D trigonometry as part of the CAPS curriculum. Approximately 30 to 45 learners will be selected, forming a sufficiently large group to permit meaningful observation of

collaborative problem-solving dynamics while remaining manageable for in-depth qualitative analysis. In addition, the Grade 10 mathematics teacher at the selected school will be included as a secondary participant, given that the teacher's facilitative practice during PBL lessons constitutes an important dimension of the phenomenon under investigation. Purposive sampling is appropriate because it ensures that all participants are directly engaged with the instructional intervention being studied and can contribute meaningful information to the research questions.

3.5 Data Collection

Data will be collected through three complementary instruments: classroom observation, semi-structured interviews, and document analysis. The use of multiple data collection methods is intended to improve the depth and comprehensiveness of the findings and to enable triangulation across data sources (Maree, 2020).

3.5.1 Classroom Observation

The researcher will conduct structured observations of Grade 10 mathematics lessons during the period in which 2D trigonometry is taught through PBL. A structured observation schedule, included in Appendix A, will be used to record observable indicators of learner engagement, problem-solving behaviour, collaborative interaction, and teacher facilitation. The schedule will capture evidence relevant to both research questions: the types of computational challenges that learners manifest during PBL activities (Research Question 1) and the ways in which PBL structures influence learners' approaches to trigonometric problem solving (Research Question 2). Observation is particularly valuable in this study because it generates direct, unmediated evidence of learner behaviour in the authentic classroom environment, complementing the self-reported data obtained through interviews.

The observer will adopt a non-participant stance during lessons, positioning himself in a location within the classroom that affords a clear view of group activities without disrupting normal lesson dynamics. Detailed field notes will be recorded during and immediately after each observed lesson, capturing verbal exchanges, written work visible on learners' desks, and the researcher's analytical reflections on emerging patterns. A minimum of six observation sessions, each corresponding to a full mathematics lesson, will be conducted.

3.5.2 *Semi-Structured Interviews*

Following the observation phase, individual semi-structured interviews will be conducted with a purposively selected subsample of approximately eight to ten learners and with the participating mathematics teacher. The interview schedule, presented in Appendix B, consists of open-ended questions designed to elicit participants' reflections on their experiences with trigonometric problem solving and with the PBL approach. Key interview themes include learners' perceptions of the challenges they encounter when computing 2D trigonometry problems, their experiences of working collaboratively in PBL groups, their views on the effectiveness of the approach, and their suggestions for further improvement. The teacher interview will explore the teacher's perspective on learners' computational difficulties, the facilitative strategies employed during PBL lessons, and the perceived impact of the approach on learner engagement and understanding.

Semi-structured interviews are appropriate for this study because they balance structure with flexibility: the predetermined schedule ensures that all interview conversations address the core research questions, while the open-ended format allows participants to raise issues that the researcher may not have anticipated and to elaborate on their responses in directions that illuminate the phenomenon in depth. All interviews will be audio-recorded with participants' consent and transcribed verbatim for analysis.

3.5.3 Document Analysis

The researcher will analyse a range of written documents produced by learners during the PBL intervention, including classwork tasks, homework assignments, formative assessment activities, and written reflections. The analysis will focus on identifying patterns in learners' computational errors, tracking changes in error frequency and type across the duration of the intervention, and examining the quality of learners' written mathematical reasoning. Document analysis provides direct evidence of the mathematical work that learners produce, offering a dimension of insight that is not readily accessible through observation or interview alone. The PBL activity sheets used during lessons will also be collected and analysed as contextual documents, enabling the researcher to evaluate the alignment between the instructional tasks and the intended learning outcomes. A document analysis guide, specifying the categories and criteria against which documents will be examined, is included in Appendix C.

3.6 Data Analysis

3.6.1 Analysis of Observation Data

Observational data — comprising field notes and the completed observation schedules — will be analysed using thematic analysis, following the six-phase process described by Braun and Clarke (2006): familiarisation with the data, initial code generation, theme construction, theme review, theme definition and naming, and report writing. Initial codes will be derived inductively from the data, capturing recurring patterns in learners' problem-solving behaviour, error types, and collaborative interactions. These codes will subsequently be organised into higher-order themes that address the research questions. For example, codes relating to learners' difficulty in selecting the appropriate trigonometric ratio may aggregate into a theme concerning the challenge of ratio identification in 2D contexts.

3.6.2 Analysis of Interview Data

Interview transcripts will be subjected to narrative analysis, a methodological approach that attends to the stories and accounts that participants construct about their experiences (Maree, 2020). Narrative analysis is particularly appropriate for interview data because it respects the temporal and contextual dimensions of participants' meaning-making, allowing the researcher to understand how learners' experiences of PBL unfold over the course of the intervention and how they relate to broader biographical and pedagogical contexts. Key analytical moves in this phase will include identifying narrative threads, noting turning points and significant events described by participants, and examining the evaluative stance that participants adopt toward their experiences. Findings from the narrative analysis will be reported through composite narrative accounts that preserve the voices of individual participants while identifying cross-cutting themes.

3.6.3 Analysis of Document Data

Learners' written work will be analysed using an error analysis framework adapted from Newman's Error Analysis (NEA) model, which classifies errors as comprehension, transformation, processing, encoding, or careless (Sahlu et al., 2025). Each learner's written solution to each trigonometric task will be coded according to this framework, enabling the researcher to produce a systematic account of the distribution and frequency of different error types before, during, and after the PBL intervention. The error analysis data will be integrated with the observational and interview data during the interpretive phase of the analysis to construct a holistic account of how PBL influences learners' computational accuracy and their awareness of their own errors.

3.7 Quality Criteria

Because this study follows a qualitative research design, the trustworthiness of the findings will be evaluated according to the four criteria proposed by Lincoln and Guba (1985): credibility, transferability, dependability, and confirmability. These criteria are now widely

accepted as the appropriate standards for assessing rigour in qualitative educational research (Ahmed, 2024; Megheirkouni & Moir, 2023).

2
1
Credibility — the qualitative analogue of internal validity — refers to the confidence with which the researcher can assert that the findings accurately represent the perspectives and experiences of the participants (Ahmed, 2024). Credibility will be enhanced through triangulation across the three data collection methods, through sustained engagement in the research site over the duration of the observation phase, and through member checking, in which selected participants will be given the opportunity to review and respond to the researcher's interpretations of their interview data before the findings are finalised.

2
38
20
Transferability — the qualitative analogue of external validity — refers to the extent to which the findings may be relevant and applicable in contexts other than the one studied (Megheirkouni & Moir, 2023). Because qualitative research does not seek statistical generalisation, transferability is achieved through the provision of a thick description of the research context, the participants, the instructional intervention, and the data collection procedures, enabling readers and future researchers to judge for themselves the extent to which the findings may resonate in comparable settings.

1
2
Dependability — the qualitative analogue of reliability — concerns the consistency and traceability of the research process over time (Ahmed, 2024). Dependability will be ensured through the maintenance of a detailed audit trail, comprising the original data, the coding schemes, the analytical memos, and the successive drafts of the thematic and narrative analyses. The audit trail will enable an independent reviewer to reconstruct the researcher's analytical journey and to assess the appropriateness of the decisions made at each stage.

17
Confirmability — the qualitative analogue of objectivity — refers to the degree to which the findings reflect the data rather than the researcher's personal biases or preferences

(Creswell & Creswell, 2018). Confirmability will be addressed through systematic reflexivity, in which the researcher maintains a reflexive journal documenting his assumptions, values, and interpretive decisions throughout the study. The use of verbatim quotations from interview transcripts and direct references to observational field notes will provide an evidentiary foundation for the interpretations offered, enabling readers to assess the degree to which the findings are grounded in the data. Daniel et al. (2024) emphasised that transparency in qualitative reporting — including explicit acknowledgement of the researcher's positionality — is a critical component of confirmability.

3.8 Ethical Considerations

The study will adhere strictly to the ethical principles governing research with human participants, with particular attention to the protections required when research involves minors. Before any data collection commences, ethical clearance will be sought from the University of Limpopo's Research Ethics Committee, and written permission will be obtained from the Limpopo Department of Education, the Mopani West District office, the school principal, and the participating teacher. These procedural safeguards ensure that the study is conducted within a framework of institutional accountability and that its potential impacts on participants and the school community have been independently reviewed.

Participation in the study will be entirely voluntary, and all participants will be provided with clear, accessible information about the purposes, procedures, and potential risks of the study before they are asked to make a decision about participation. Because the learner participants are minors, written informed consent will be sought from their parents or legal guardians (Appendix D), and written assent will be sought from the learners themselves (Appendix E). Both consent and assent documents will be prepared in language appropriate to the intended audience. Participants will be explicitly informed that they retain the right to

withdraw from the study at any point, without any adverse consequences for themselves or their academic standing (Creswell & Creswell, 2018).

The confidentiality and anonymity of all participants will be maintained throughout the research process and in all published outputs. No real names will be used; instead, alphanumeric codes will be assigned to each participant, and the name of the school will be replaced with a pseudonym in all research documents. Audio recordings and transcripts will be stored in a password-protected digital repository accessible only to the researcher and the supervisor, and will be destroyed five years after the completion of the study. The researcher will ensure that data collection and analysis activities do not cause physical, psychological, or social harm to any participant (Cohen et al., 2018). The study will also be conducted in a manner that does not disrupt the normal teaching and learning programme of the school, with observations and interviews scheduled at times that are agreed upon in advance with the teacher and the school management.

3.9 Conclusion

This chapter has detailed the research methodology that will guide the study. The interpretivist paradigm and qualitative case study design are well suited to the exploratory and context-sensitive nature of the research questions. The combination of classroom observation, semi-structured interviews, and document analysis will generate a multi-perspectival data set from which a rich and credible account of the relationship between PBL and Grade 10 learners' trigonometric computation can be constructed. The quality criteria and ethical safeguards outlined above ensure that the study will be conducted with rigour and with appropriate respect for the rights and wellbeing of all participants.

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APPENDICES

Appendix A: Classroom Observation Schedule

The following observation schedule will be used during each classroom observation session.

The researcher will observe Grade 10 mathematics lessons during which 2D trigonometry problems are taught through Problem-Based Learning. For each session, the researcher will record the date, the lesson topic, the number of learners present, and detailed observational notes organised according to the following focus areas.

Focus Area 1: Learner Engagement. The researcher will note the extent to which learners participate actively in PBL group discussions, the nature and quality of contributions made by individual learners and groups, and the strategies learners employ when they encounter difficulty.

Focus Area 2: Problem-Solving Behaviour. The researcher will document the approaches learners adopt when analysing trigonometric problems, including their ability to identify relevant information, select appropriate trigonometric ratios, execute calculations, and verify their answers.

Focus Area 3: Learner-Learner and Learner-Teacher Interaction. The researcher will record episodes of peer scaffolding, peer explanation, and teacher facilitation, noting the mathematical content of these interactions and their apparent impact on learner understanding.

Focus Area 4: Error Patterns. The researcher will make notes of observable errors and misconceptions, categorised according to the Newman Error Analysis framework, and will record the contexts in which these errors arise and the strategies (if any) that learners employ to detect and correct them.

Space for field notes and reflective commentary will be provided on the reverse of each completed schedule.

Appendix B: Interview Schedule

B1: Learner Interview Schedule

The following questions will guide interviews with Grade 10 learner participants. Probing questions will be added as appropriate during each interview to clarify and extend participants' responses.

1. Can you describe what you find most difficult about solving two-dimensional trigonometry problems? What specifically makes those parts hard for you?
2. When you are working in a group during a PBL mathematics lesson, how does the group activity help you — or not help you — to understand trigonometry better?
3. Walk me through how you approach a 2D trigonometry problem from start to finish. What steps do you follow, and where do you sometimes get stuck?
4. How do you feel about learning mathematics through group problem-solving activities compared to having the teacher explain everything at the front?
5. What changes do you think could be made to the way trigonometry is taught in your class to help you and your classmates understand it better?

B2: Teacher Interview Schedule

1. In your experience teaching Grade 10 mathematics, what are the most common challenges your learners encounter when working through 2D trigonometry problems?

2. How do you typically facilitate PBL lessons in your classroom, and what facilitation strategies do you find most effective for supporting learners during trigonometry tasks?

3. In what ways, if any, have you observed learners' computational accuracy or conceptual understanding in trigonometry change since you introduced PBL?

4. What challenges have you encountered in implementing PBL for trigonometry, and how have you addressed them?

Appendix C: Document Analysis Guide

The document analysis guide will be applied to all learner-produced written work collected during the study. Documents will be analysed according to the following categories, derived from the Newman Error Analysis framework and from the specific demands of 2D trigonometry computation.

Category 1 — Comprehension accuracy: Did the learner correctly identify the given information, the unknown quantity, and the type of triangle involved in the problem?

Category 2 — Ratio selection: Did the learner select the appropriate trigonometric ratio (sine, cosine, or tangent) for the given configuration of the triangle?

Category 3 — Transformation accuracy: Did the learner correctly rearrange the trigonometric equation to isolate the unknown?

Category 4 — Calculation accuracy: Did the learner execute the arithmetic and calculator-based calculations correctly?

Category 5 — Encoding accuracy: Did the learner express the final answer in the correct form and with the appropriate units?

For each document, the researcher will assign a code indicating the presence or absence of errors in each category and will record brief analytical notes explaining the nature of any errors identified. These codes will subsequently be used to generate frequency distributions of error types across the learner group and to track changes in error patterns across the duration of the PBL intervention.

Appendix D: Parent/Guardian Consent Form

UNIVERSITY OF LIMPOPO FACULTY OF HUMANITIES

INFORMED CONSENT FORM FOR PARENTS/GUARDIANS

Research Title: Learners' Computation of Trigonometry 2D Problems Using Problem-Based Learning at a School in Lepelle 04 Circuit, Mopani West District, Limpopo Province, South Africa.

Dear Parent/Guardian,

Your child is invited to participate in a research study that investigates how Problem-Based Learning can support Grade 10 mathematics learners in solving two-dimensional trigonometry problems. The study is being conducted by Mr MD [REDACTED] (Student Number: [REDACTED]) under the supervision of Prof ZB Dhlamini at the University of Limpopo, as part of the requirements for the degree of Honours in Mathematics Education.

Participation in the study is entirely voluntary. Your child's participation may involve taking part in classroom activities, group problem-solving tasks, and an individual interview. All information collected will be used solely for academic purposes and will be treated with strict confidentiality. Your child's name and the name of the school will not appear in any research reports or publications; pseudonyms and codes will be used instead. You and your child retain the right to withdraw from the study at any time and without any penalty.

If you agree for your child to participate in this study, please complete and sign the section below and return this form to the class teacher.

I, _____, parent/guardian of _____, hereby give permission for my child to participate in the above-mentioned research study. I understand that participation is voluntary and that my child may withdraw at any time.

Signature: _____ Date: _____

Contact Number: _____

Appendix E: Learner Assent Form

LEARNER ASSENT FORM

Research Title: Learners' Computation of Trigonometry 2D Problems Using Problem-Based Learning at a School in Lepelle 04 Circuit, Mopani West District, Limpopo Province, South Africa.

Dear Learner,

 45 You are invited to take part in a research study about learning trigonometry through Problem-Based Learning activities. The study aims to help us understand the challenges that Grade 10 learners face when solving two-dimensional trigonometry problems and how PBL can support learning. You may be asked to participate in classroom activities, group problem-solving tasks, and a short interview.

 42 Your participation is your own choice. You may choose not to participate or to stop participating at any point without any negative consequences. Everything you tell us and all the work you produce will be kept private. Your name will not appear in any reports or publications arising from this study.

If you understand the information above and agree to take part, please sign below.

Learner Name: _____

Signature: _____ Date: _____